Automating Legal Reasoning with Generative AI: A Framework for Judicial Decision Support

FirstName Surname†
Department Name
Institution/University Name
City State Country
email@email.com

FirstName Surname
Department Name
Institution/University Name
City State Country
email@email.com

FirstName Surname
Department Name
Institution/University Name
City State Country
email@email.com

ABSTRACT

This article presents a framework for automating legal reasoning in judicial decisions using LLMs. It combines advanced techniques, including Chain-of-Density prompting, the CO-STAR methodology, and the FIRAC model, to analyze claims, requests, and evidence, generating structured and coherent reasoning. Evaluated on first-instance cases from a Brazilian State Court of Justice, the framework was assessed using ROUGE, METEOR, CHRF, TER, ALIGNSCORE, and BERTSCORE metrics. While lexical metrics revealed challenges in abstractive text generation, semantic metrics showed strong alignment with reference texts, confirming the framework's ability to preserve meaning. The results highlight its potential to streamline judicial workflows, enhance consistency, and improve transparency.

CCS CONCEPTS

• Applied computing → Law; • Computing methodologies → Artificial intelligence → Natural language generation

KEYWORDS

Generative Artificial Intelligence (GAI), Legal Reasoning Automation, Judicial Decision-Making, and Large Language Models (LLMs)

1 Introduction

Large Language Models (LLMs)[27] demonstrate significant potential in judicial applications, including legal claim analysis, evidence evaluation, and the generation of structured legal reasoning [11],[28]. Drafting judicial decisions requires a complex interplay of legal expertise, factual assessment, and argument interpretation. As key instruments in dispute resolution, judicial decisions must be legally sound, adhere to procedural norms, and conclude the cognitive phase of litigation while ensuring enforcement measures.

Formulating judicial decisions, particularly on core legal issues, is an intellectually demanding task that involves coordinating extensive case files, analyzing legal norms, and balancing jurisprudence, principles, and doctrinal interpretations. Judges, judicial assistants, and clerks navigate vast and complex information, making it challenging to produce well-reasoned decisions efficiently. The volume of relevant evidence, including testimonies and expert reports, further complicates this process.

The National Council of Justice (CNJ) Resolution No. 332/2020 establishes guidelines for the ethical and transparent use of AI in the judiciary, emphasizing the need for reasoned decisions and the protection of fundamental rights. However, integrating evidentiary elements with legal arguments in a structured manner remains a challenge. This study addresses this gap by proposing a framework that leverages LLMs alongside robust legal reasoning methods, enhancing the coverage and coherence of judicial reasoning. A key feature is the construction of a case graph, enabling verification of evidence completeness and provenance.

Implemented within the SARA system, the framework was evaluated by judges from a Brazilian State Court of Justice using a gold-standard collection of first-instance decisions. The experimental analysis employed ALIGNSCORE-Embeddings[23] and BERTSCORE[24] to assess semantic preservation and abstractive text generation. Results indicated moderate to strong performance in generating legal reasoning using GPT-4, demonstrating the framework's potential to enhance judicial workflows and decision quality.

2 Related Work

The application of Artificial Intelligence (AI) in the legal domain has advanced significantly, particularly in the analysis of evidence and the generation of texts related to legal reasoning. Recent studies have highlighted how AI can enhance accuracy and efficiency in the management of judicial data. For example, [12] provides a comprehensive analysis of the transformative impact of AI on legal research and forensic practices, emphasizing its ability to process large volumes of data and identify patterns in digital and biometric evidence. These advancements facilitate the identification of relevant evidence for resolving complex cases.

In the field of forensic analysis, [13] examines the integration of AI into forensic analysis, proposing criteria for incorporating technologies into forensic systems. The authors emphasize the necessity of continuous validation of AI-generated results, with the PDCA cycle (Plan, Do, Check, Act) identified as an effective model for ensuring the reliability of such systems. This approach underscores the importance of constant evaluation to maintain the adequacy and accuracy of forensic outcomes.

Furthermore, [14] explores the application of deep learning technologies, such as convolutional neural networks (CNNs) and natural language processing (NLP), for analyzing judicial evidence. This research demonstrates how hybrid models can enhance accuracy in identifying evidence and establishing its connection to the arguments presented by the parties.

Studies such as [15] address the use of AI for the automated evaluation of evidence in international judicial proceedings, focusing on admissibility and relevance within the context of diverse legal systems. The study highlights the importance of transparent and ethical systems to ensure compliance with legal standards and the protection of the rights of the parties involved.

Another critical aspect is the use of predictive modeling in judicial decision-making, as detailed in [16]. This study examines 13 deep learning models applied to predicting legal outcomes based on case files and highlights improvements in legal document classification, particularly in overburdened judicial systems. However, studies such as [17] raise concerns about the ethical challenges posed by integrating AI into evidence analysis and legal reasoning, including issues of algorithmic bias, data privacy, and the interpretability of AI-generated results. These challenges are pivotal to ensuring that AI technologies adhere to legal principles and maintain impartiality.

3 Framework for Generating the Legal Reasoning of Judicial Decisions

This section introduces a framework based on Generative Large Language Models (LLMs), such as GPT-4 and Llama 3, designed to generate legal reasoning for judicial decisions (see Figure 1). The framework is composed of sequential modules that handle pre-processing of case documents, analysis of claims and arguments presented by the parties, evaluation of the submitted evidence, and, ultimately, the generation of the legal reasoning. Each module employs robust prompt engineering, which incorporates evidentiary elements and the parties' claims, ensuring that all relevant arguments are analyzed and that the reasoning is clear, transparent, and legally substantiated.

In general, the workflow begins with the collection of relevant legal documents, including petitions, defenses, expert reports, and other materials related to the case, along with the evidence submitted by the parties. In Module_1 (Pre-Processing), these documents and evidence are analyzed using image processing models and optical character recognition (OCR) tools. This step aims to extract content by converting images and scanned documents into machine-readable text. In Module_2, the textual content of documents such as the initial petition, defenses, and replies is analyzed by the LLM to extract and structure the claims and arguments of the parties. Module_3 focuses on the analysis of evidence using refined prompts submitted to the LLM, which results in the organization of evidence, verification of its provenance, and linkage of evidence to the claims and arguments. Finally, in Module 4, the outputs of the previous modules—combined with the case report and other case-related documents—are submitted via prompt to the LLM to generate the legal reasoning grounding the judicial decision.

The subsequent sections provide a detailed description of each module, its stages, and the prompt engineering techniques applied to ensure the accuracy and coherence of the framework.

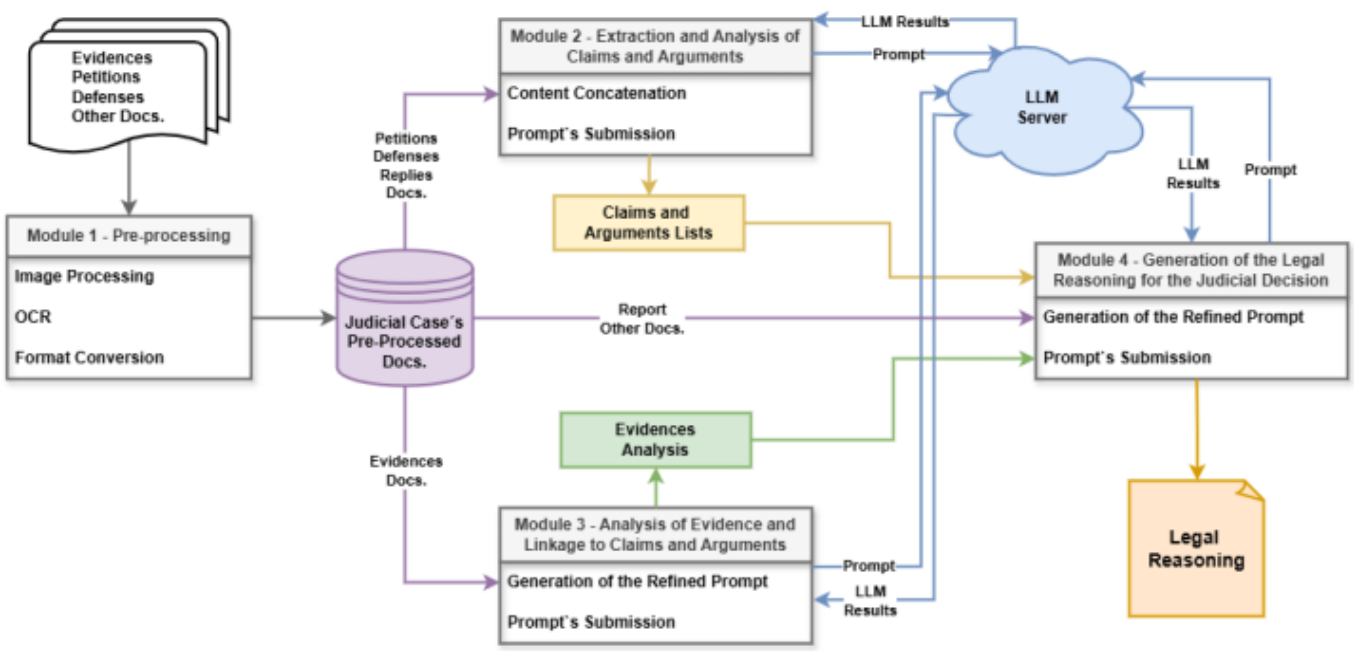

Figure 1: Framework for generating the legal reasoning of judicial decisions.

Module_1 - Pre-processing

Pre-processing covers all the necessary actions to transform raw documents, images, and information from the case into a comprehensible and organized format, enabling LLMs to perform the analysis and generation tasks as specified in each prompt. Many case documents (PDFs, handwritten documents, older scans) are often in formats that hinder direct analysis by AI tools. Converting these documents into JPEG/PNG images, followed by Base64 encoding, allows the LLM to read and interpret the content using specific instructions.

Module 2 - Extraction and Analysis of Claims and Arguments

This structured format for claims and requests facilitates subsequent integration and retrieval of each party's submissions and the associated legal arguments, enabling their linkage to the evidence presented by each party and their relationship to the reasoning in the legal decision. Figure 2 illustrates an example of the output from the prompt analyzing claims and requests in a judicial case, showing six claims and seven requests from the plaintiff. In the example, the following claim was extracted from the plaintiff: <ap1> = "The claimant did not request the service contract and was a victim of fraud."

Module 3 - Analysis of Evidence and Connection to Claims and Arguments

The objective of this module is to analyze the content of the evidence submitted in the case, alongside the outputs from the previous phase (a structured list of claims and requests), utilizing a refined prompt (the full version of the prompt is provided in Appendix 2). At this stage, the LLM is responsible not only for describing the evidence but also for establishing explicit links between the evidence and specific claims or requests made by the parties.

```
Requests from the Parties:
List 1: Allegations of the Claimant
1. Allegation that the claimant did not request the service contract and was a victim of fraud `<ap1>`.
2. Declaration that there was no consent or signature for the contract, with indications of forgery `<ap2>`.
3. Statement that the received amount was transferred without authorization or prior knowledge `<ap3>`.
4. Argument that the deductions made directly impact the claimant's sole source of income `<ap4>`.
5. Allegation that the institution did not take proper precautions to prevent irregularities `<ap5>`.
6. Declaration that there was an attempt at extrajudicial resolution, but without success `<ap6>`.
List 2: Requests of the Claimant
1. Request for the granting of free legal aid `<pa1>`.
2. Request for procedural priority, based on specific legal requirements `<pa2>`.
3. Request for preliminary injunction to immediately suspend charges and negative credit reporting `<pa3>`.
4. Request for a declaration of non-existence of the legal relationship and the amounts charged `<pa4>`.
5. Claim for moral damages, with the amount to be determined by the court `<pa5>`.
6. Request for double reimbursement of improperly charged amounts `<pa6>`.
7. Request for the defendant to be ordered to pay costs and attorney's fees `<pa7>`.
```

Figure 2 : An example of output from the prompt analyzing claims and requests in a judicial case.

The goal of this module is to identify, within the set of documents (e.g., initial petitions, defenses, replies, etc.), the claims and arguments presented by both the plaintiff and the defendant, as well as any additional requests made by other entities (such as judges), if applicable.

Step 1: Content Concatenation

In cases where multiple documents exist, their content is merged into a single text to enable comprehensive analysis by the model.

Step 2: Submission of a Specialized Prompt

A specialized prompt (Appendix 1 contains the complete version of the prompt) is submitted to the LLM, requesting the extraction of the claims and arguments of the plaintiff, defendant, and other entities, while preserving clarity and fidelity to the original text. The output is structured into lists of claims and arguments according to the following template:

- Claims by the Plaintiff: <ap1>, <ap2>, ...
- Requests by the Plaintiff: <pa1>, <pa2>, ...
- Claims by the Defendant: <ar1>, <ar2>, ...
- Requests by the Defendant: <pr1>, <pr2>, ...
- Additional or Relevant Requests: <op1>, <op2>, ...

Step 1: Generation of a refined prompt

Generation of a refined prompt requesting the analysis of evidence contained in {***filename***}, considering the lists of claims and requests extracted from the previous module ({***pedidos_content***}), and the extraction of mandatory elements according to the following template:

> *Analyze the image containing evidence from a judicial case in: '**{filename}**' and assess it based on the following claims and requests **{pedidos_content}** (...) Step 1: ... Step 3: Develop the analysis of the evidence with the following mandatory elements: **Title:** Extract the name of the document, if available. **Subtitle:** Provide a descriptive subtitle that synthesizes the subject of the evidence. **Date:** Extract and record any*

relevant date found in the image (if present). ***Page:*** *Mention the page number as it appears in the file name (e.g., Page(99)).* ***Sheet:*** *Mention the sheet number contained in the image (e.g., fls(99)).* ***Produced by:*** *Identify who produced the evidence.* ***Requested by:*** *Identify who requested the evidence.* ***Objective:*** *Specify which claims or requests the evidence addresses.* ***Outcome of the evidence:*** *Summarize the findings of the evidence.*

Step 2: Submission of the Refined Prompt

The refined prompt is submitted to the LLM and the result of the evidence analysis is registered. Figure 3 illustrates the result of the analysis of the evidence "PROVA_DOCUMENTAL_PARTE_autor_page2.jpg," including the mandatory elements such as title, subtitle, date, produced by, requested by, among others. In this example, the **Objective** of the evidence presented by the plaintiff in the judicial case was "to prove the attempt to resolve the issue administratively by the petitioner."

- The refinement of the evidence analysis prompt, integrating the contextual nuances of the claims and requests presented by the parties.

Module 4 - Generation of the Legal Reasoning for the Judicial Decision

This module is responsible for generating the legal reasoning text that supports the judicial decision, using the prompt referred to as "Prompt_Fundamentacao_<(omitted by blind review)>," briefly outlined in Table 1. The design of this prompt incorporates an innovative combination of the CO-STAR framework [26], the Chain-of-Density (CoD) prompt engineering technique [25], and the FIRAC method [30]. This approach aims to enhance the precision and depth of legal reasoning by producing well-structured arguments directly connected to case elements, such as evidence, claims, and requests. Additionally, the prompt is designed to comprehensively integrate decisional reasoning, ensuring detailed consideration of all evidence, claims, and requests presented during the proceedings.

```
Requests from the Parties:
List 1: Allegations of the Claimant
1. Allegation that the claimant did not request the service contract and was a victim of fraud `<ap1>`.
2. Declaration that there was no consent or signature for the contract, with indications of forgery `<ap2>`.
3. Statement that the received amount was transferred without authorization or prior knowledge `<ap3>`.
4. Argument that the deductions made directly impact the claimant's sole source of income `<ap4>`.
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List 2: Requests of the Claimant
1. Request for the granting of free legal aid `<pa1>`.
2. Request for procedural priority, based on specific legal requirements `<pa2>`.
3. Request for preliminary injunction to immediately suspend charges and negative credit reporting `<pa3>`.
4. Request for a declaration of non-existence of the legal relationship and the amounts charged `<pa4>`.
5. Claim for moral damages, with the amount to be determined by the court `<pa5>`.
6. Request for double reimbursement of improperly charged amounts `<pa6>`.
7. Request for the defendant to be ordered to pay costs and attorney's fees `<pa7>`.
```

Figure 3: An example of the result of the analysis of the evidence

By explicitly incorporating claims and requests into the prompt for evidence analysis, the LLM is provided with refined information to reason about the context and relevance of each piece of evidence, resulting in more precise analyses. This strategy enhances the coherence between evidence and claims, maximizing the potential for robust and reliable legal reasoning.

Upon completing the execution of the modules outlined above, a comprehensive dataset and analytical framework are prepared for the subsequent phase of drafting the judicial decision's reasoning:

- The processing of evidence images and documents, ensuring their conversion into a format interpretable by Large Language Models (LLMs);
- The systematic organization of all claims and requests submitted by the parties;

Table 1 provides a structured overview of the prompt based on the CO-STAR methodology, organized into the following sections: CONTEXT, OBJECTIVE, STYLE, TONE, AUDIENCE, RESPONSE, and GUIDELINES. The prompt includes the following parameters, which are configured at the time of its instantiation within the framework: ***conteudo_pedidos***, containing the list of claims and requests (see Figure 2); ***conteudo_documentos***, which includes all case documents, such as the initial petition, defenses, closing arguments, and the case report; ***conteudo_provas***, encompassing the analysis of evidence detailed in Module 3; and ***verdict,*** specifying the judgment outcome as granted (G), denied (D), or partially granted (PG).

As described in Module_2 - Extraction and Analysis of Claims and Arguments, the tags associated with each request or claim play a crucial role in legal reasoning, as they allow linking them

to the evidence and arguments presented. The prompt instructs the LLM to delimit specific sections of the text that refer to the claims and requests, using tags such as <inicio apN> and <fim apN>. This enables the model to explicitly structure and connect legal reasoning to case evidence, facilitating audits and reviews while ensuring the coherence and accuracy required for legal analysis.

Figure 5 presents the generation of the legal reasoning for a case. In the example, the tags <inicio ap1> / <fim ap1> delimit the reference to the first claim made by the claimant, which alleges being a victim of fraud in the execution of a payroll loan contract. Similarly, the tags <inicio ap2> / <fim ap2> mark the reference to the second claim of the claimant, questioning the authenticity of the signature on the contract. In summary, the presence of tags is not limited to mere formal labeling: they play a practical role by enabling easy review of the coverage of requests, claims, and evidence in the case by legal advisors and judges.

Version: Prompt_Fundamentacao_<(omitted by blind review)>

CONTEXT

. You will receive a list of requests and claims, along with an analysis of the evidence, resulting from the pre-processing of the evidence submitted in a judicial case (...), as well as the **verdict** (GRANTED, DENIED, or PARTIALLY GRANTED).

OBJECTIVE

The objective is to assist in drafting the reasoning of a judicial decision, prioritizing the review and validation of the evidence. That is, the foundation must be based on the analysis and reference to the evidence (...), using the FIRAC method (Facts, Issue, Rule, Analysis, Conclusion) as the judgment structure.
(...)
If any of this information is not available in the documents, record its absence clearly in the reasoning.

STYLE

The text should be narrative, formal, and analytical, with explicit references to the evidence and legal norms. Mention the pages (p.) where the information is found, maintaining clarity and precision.

TONE

The tone should be assertive and informative, ensuring that all information is presented with precision and relevance, without expressing personal opinions.

AUDIENCE

Judges and legal professionals involved in the case.

RESPONSE

The reasoning must be written in narrative form, without adjectives. It must respect the informed **verdict** (GRANTED, DENIED, or PARTIALLY GRANTED) and be based exclusively on the evidence and procedural documents.
TEMPLATE:
MODEL 1 – Decision granting claims without preliminary issues/prejudicial matters:
REASONING

The central point of the controversy is to decide whether [CENTRAL ISSUE OF THE CONTROVERSY]. In other words, [CENTRAL ISSUE FROM ANOTHER PERSPECTIVE].
(...)

GUIDELINES

1. Evidence as Essential Elements with References to Pages (p.):
Step 1.1 - Include and analyze all relevant evidence supporting the legal reasoning in accordance with the **VERDICT**, ensuring that each piece of evidence cited mentions the corresponding page (p.) for ease of reference and traceability.
Step 1.3 - Verification: Confirm that all evidence has been cited and that all evidence is clearly associated with the corresponding legal argument, that the pages (p.) are correctly mentioned, and that they have been marked with the appropriate TAGs. Review whether each evidence citation is complete and clear (...).
Step 1.4 - Verify once more and check whether, according to the **VERDICT**, the requests and claims have been addressed: (...)
Repeat steps 1.2 to 1.4 five times.

2. Density and Precision:
Step 2.1: Identify 1 to 5 missing informative entities (...).
Repeat step 2.2 five times.
Step 2.2: Write a new and denser summary, of similar length, that covers all entities and details from the previous summary, as well as the missing entities (...).

3. Clarity and Accessibility: (...)
4. Essentiality and Fidelity to the Report: (...)
5. Complete and Detailed Structure: (...)
Verification: Review the text to ensure that all arguments are properly explored. Compare with a list of expected legal issues to ensure full coverage and the necessary depth.

6. Self-Sufficient Final Content:
Verification: Perform a final reading without consulting the report to confirm that the text is self-sufficient.

7. Strict Formatting According to the TEMPLATE:
(...)
After ensuring the TEMPLATE's application, strictly follow the annotation rules below:
Step 7.1 - Enclose all content with a main tag <content> to indicate the beginning and end of the set.
Step 7.2 - Mandatory identification of each request and claim in the generated text with the respective TAG.
Step 7.3 - Include a start and end TAG indicating each request or claim represented. Example: <start rq1> <end rq1>.
Repeat steps 7.2 and 7.3 five times to ensure that the entire text has been identified with TAGs (...).

8. Verification: After drafting, compare the content with the TEMPLATE and examples to ensure exact correspondence in structure. Confirm that all formatting details (such as form, style, and structure) are identical to the provided model..

Table 1 - Prompt for generation of legal reasoning for judicial decisions

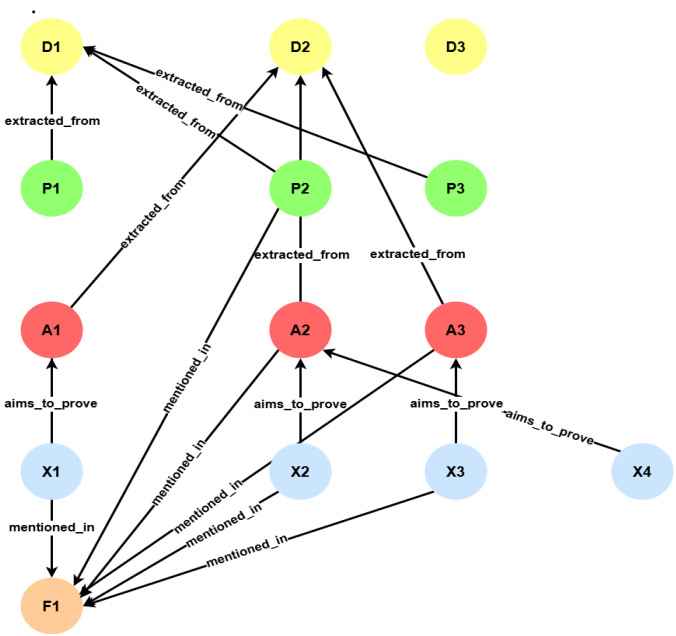
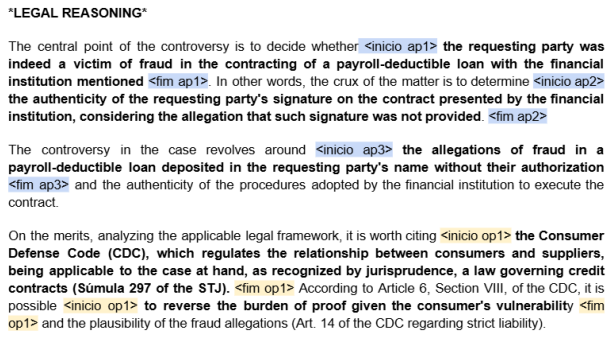


Figure 4: Hierarchical graph representation of legal relationships.



LEGAL REASONING

The central point of the controversy is to decide whether <inicio ap1> **the requesting party was indeed a victim of fraud in the contracting of a payroll-deductible loan with the financial institution mentioned** <fim ap1>. In other words, the crux of the matter is to determine <inicio ap2> **the authenticity of the requesting party's signature on the contract presented by the financial institution, considering the allegation that such signature was not provided.** <fim ap2>

The controversy in the case revolves around <inicio ap3> **the allegations of fraud in a payroll-deductible loan deposited in the requesting party's name without their authorization** <fim ap3> and the authenticity of the procedures adopted by the financial institution to execute the contract.

On the merits, analyzing the applicable legal framework, it is worth citing <inicio op1> **the Consumer Defense Code (CDC), which regulates the relationship between consumers and suppliers, being applicable to the case at hand, as recognized by jurisprudence, a law governing credit contracts (Súmula 297 of the STJ).** <fim op1> According to Article 6, Section VIII, of the CDC, it is possible <inicio op1> **to reverse the burden of proof given the consumer's vulnerability** <fim op1> and the plausibility of the fraud allegations (Art. 14 of the CDC regarding strict liability).

Figure 5: Legal reasoning with tags for claims and arguments.

At the conclusion of the entire process, the framework is capable of generating a graph that represents the relationships between the elements of the case and the legal reasoning. Figure 4 illustrates a graph generated for a hypothetical judicial case, structured as follows:

- **Documents:** The graph contains the judicial documents D1, D2 e D3.
- **Claims:** P1, P2 and P3 were extracted from document D1, represented by the relations:
  - P1 -> extracted_from -> D1
  - P1 -> extracted_from -> D1
  - P2 -> extracted_from -> D1
  - P3 -> extracted_from -> D1
- **Arguments:** A1, A2 e A3 were extracted from document D2, represented by the relations:
  - A1 -> extracted_from -> D2
  - A2 -> extracted_from -> D2
  - A3 -> extracted_from -> D2
- **Evidences:** X1, X2, X3 e X4 presented by the plaintiff and defendant, with the following objectives:
  - X1 -> aims_to_prove -> A1
  - X2 -> aims_to_prove -> A2
  - X3 -> aims_to_prove -> A3
  - X4 -> aims_to_prove -> A2
- **Legal Reasoning:** F1 mentions the following elements:
  - P2 -> mentioned_in -> F1
  - P3 -> mentioned_in -> F1
  - A2 -> mentioned_in -> F1
  - A3 -> mentioned_in -> F1
  - X1 -> mentioned_in -> F1
  - X2 -> mentioned_in -> F1
  - X3 -> mentioned_in -> F1

In this example, certain elements, such as request P1, claim A1, and evidence X4, were not referenced in the reasoning generated by the framework. This informational gap is critical for analysis and review by legal professionals, such as judges and clerks, helping to prevent procedural inconsistencies, such as motions for clarification. Additionally, our framework lays the groundwork for integration with Ontologies and Knowledge Graphs within the field of Legal AI, fostering greater explainability and transparency in judicial decision-making processes. We argue that this structure enables the tracking, verification, and comprehension of how case elements and evidence were utilized in the reasoning, providing a robust foundation for more consistent and well-justified decisions.

4. Implementation and Experimental Evaluation

4.1 Implementation in SARA

The proposed framework was implemented in the Sistema de Análise e Resumo de Ações (SARA <(omitted by blind review)>) [29], utilizing the GPT-4 model [31], accessed via the OpenAI API with default hyperparameter settings. The SARA interface (Figure 6) was designed to intuitively and functionally present the legal reasoning and references to evidence, claims, and requests highlighted in the text, employing color coding to facilitate the identification of key elements in the case.

On the left side of the interface, the generated legal reasoning is displayed, with emphasis on references to evidence and legal norms. The text features interactive colors directly linked to corresponding lists on the right side of the screen. Each color represents a specific type of element (claims, requests, evidence), simplifying access and enabling quick verification of the mentioned items. On the right side, structured lists of the case's fundamental elements are displayed. These lists include the claims and requests of each party (plaintiff and defendant), as well as the evidence produced, ensuring that all items are considered in the generated legal reasoning.

The interface emphasizes the system's functionality by connecting legal analyses to the original documents, promoting

clear, traceable, and consistent legal reasoning. Furthermore, the highlighted elements are interactive: clicking on any reference opens a corresponding tab displaying the related content, such as

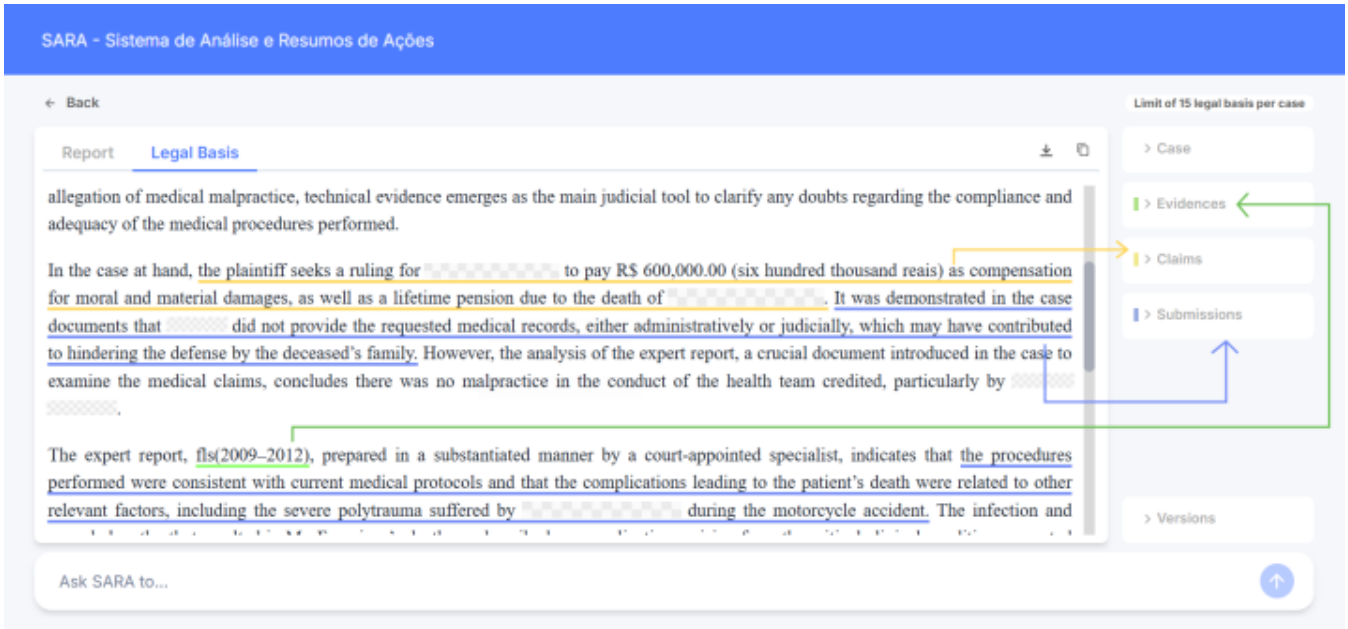


Figure 6 : Legal reasoning with visual highlights in the text for evidence, claims and arguments.

the evidence page or the specific item from the claims or requests list (see Appendix 3).

4.2 Experimental Evaluation

The experimental evaluation involved generating legal reasoning for a gold-standard collection of five first-instance judicial cases from the Court of Justice <(omitted by blind review)>, Brazil. These cases were selected by the civil courts participating in the project. The legal reasonings generated by the framework were compared to those authored by the judges, using the following metrics: ROUGE [10], METEOR [20], CHRF [21], TER [22], ALIGNSCORE [23], and BERTSCORE [24].

Table 2 presents the evaluation results based on the ROUGE metrics (F1-Score), including ROUGE-1, ROUGE-2, and ROUGE-L, which are widely used in the automatic evaluation of generated texts relative to reference texts. These metrics provide a quantitative analysis of lexical and structural overlap. However, their application to abstractive text generation, as in this study, requires careful consideration, as these metrics were originally designed for extractive text summarization, which exhibits higher literal overlap. For the generated legal reasoning, ROUGE-1 scores ranged from 0.18 to 0.32, with an average of 0.24, indicating low alignment with unigrams from the reference texts. ROUGE-2, which measures the overlap of consecutive word pairs (bigrams), ranged from 0.08 to 0.14, with even lower values. Such low scores are expected in abstractive text generation, as bigrams rely on similar sentence structures between the generated and reference texts. ROUGE-L, which evaluates the overlap of the longest common subsequences (LCS), presented scores ranging from 0.08 to 0.13. Low scores across these metrics suggest that the generated text may be fragmented and disorganized concerning the logical progression of facts.

Additional metrics were employed to quantitatively evaluate the proposed legal reasoning generation model. Table 3 and 4 presents the results of the METEOR, CHRF, TER, ALIGNSCORE, and BERTSCORE metrics for the judicial cases included in this evaluation.

File_Name	ROUGE-1 F1	ROUGE-2 F1	ROUGE-L F1
0255892-30.2021	0,318142	0,149452	0,130372
0207225-76.2022	0,247849	0,120462	0,109946
0181397-83.2019	0,180695	0,082181	0,084307
0154724-53.2019	0,179250	0,083401	0,085823
0131659-97.2017	0,253450	0,100981	0,109659

Table 2. Evaluation of Generated Legal Reasonings According to ROUGE-1, ROUGE-2, and ROUGE-L Metrics (F1-Score)

The METEOR metric evaluates the similarity between the generated text and the reference text, accounting for synonyms and grammatical inflections. Scores ranged from 0.05 to 0.10, reflecting limited performance typical of abstractive text generation. While METEOR is effective at assessing the preservation of synonyms and inflections, it may underestimate the quality of texts that utilize more advanced paraphrasing, which is common in abstractive approaches. The CHRF metric assesses textual similarity based on character-level n-grams, making it more sensitive to subtle changes in text structure.

Scores ranged from 12.11 to 22.78, indicating significant variations in text alignment. CHRF's sensitivity to grammar and

style variations means that lower scores may reflect the abstractive nature of the generated text, which rephrases and reorganizes information rather than replicating the structure of the reference text. The TER metric measures the number of edits (insertions, deletions, substitutions) required to transform the generated text into the reference text. Scores fell within the range of 95.05 to 97.54, indicating a high editing effort. TER is particularly useful for evaluating structural fidelity but tends to penalize the reformulations typical of abstractive approaches. High scores are expected for models that prioritize flexibility in text generation.

File_Name	METEOR	CHRF	TER
0255892-30.2021	0,104104	22,780474	95,054257
0207225-76.2022	0,070903	15,520626	95,940410
0181397-83.2019	0,050273	12,304779	97,540555
0154724-53.2019	0,052666	12,118569	96,948783
0131659-97.2017	0,073416	18,581456	96,441901

Table 3. Evaluation of Generated Legal Reasonings According to METEOR, CHRF, and TER Metrics

File Name	ALIGNSCORE _Embeddings	ALIGNSCORE_Lexical	BERTSCORE _F1
0255892-30.2021	0,496633	0,130789	0,716260
0207225-76.2022	0,359828	0,115365	0,672584
0181397-83.2019	0,251885	0,097815	0,657404
0154724-53.2019	0,325663	0,089226	0,723032
0131659-97.2017	0,409137	0,103349	0,669207

Table 4. Evaluation of Generated Legal Reasonings According to ALIGNSCORE and BERTSCORE Metrics

The ALIGNSCORE metric was employed in this evaluation with two variants: Embeddings (cosine similarity between semantic embeddings) and Lexical (lexical similarity based on the Jaccard index). For the ALIGNSCORE-Embeddings variant, values ranged from 0.25 to 0.49, indicating low to moderate semantic similarity with the reference texts. In the ALIGNSCORE-Lexical variant, values ranged from 0.08 to 0.13, suggesting significant divergence in terms of the words used in the generated texts compared to the references. The BERTSCORE metric showed a range of 0.65 to 0.72, with an average score of 0.70. This metric evaluates semantic similarity between texts by leveraging contextual embeddings and is particularly suitable for assessing abstractive text generation, as it measures the preservation of meaning independently of textual structure. The results suggest that the model effectively captures the overall meaning of the text, despite potential stylistic differences, indicating that the generated texts preserved the semantics of the reference texts.

Overall, in this initial quantitative evaluation, metrics such as METEOR, CHRF, TER, and ALIGNSCORE-Lexical—which heavily penalized reformulations and paraphrasing, common in abstractive summaries—indicated unsatisfactory performance for the proposed model. In contrast, ALIGNSCORE-Embeddings and BERTSCORE, which better capture semantic preservation and are more suitable for evaluating abstractive generation in legal scenarios, demonstrated moderate to good performance for the legal reasonings generated under the proposed framework utilizing the GPT-4 LLM.

In an ad-hoc qualitative evaluation conducted by judges from the <(omitted by blind review)>, all participants unanimously rated the generated reasoning as satisfactory for supporting judicial decisions. However, they highlighted the utility of the extracted relationships between claims, arguments, and evidence, as well as the references to these elements within the legal reasoning, enabling an analysis of the coverage of case elements. In this context, we analyzed the quantity of claims, arguments, and evidence in each case and the coverage percentage (%C) within the legal reasoning. Table 5 presents these statistics. The analysis reveals that, in cases ruled as fully procedent (G) or partially procedent (PG), there is greater utilization of the plaintiff's arguments and claims, reflecting their central role in the legal reasoning. Conversely, in non-procedent cases (D), the focus shifts toward the defendant's arguments, indicating that the decision was primarily based on refuting the plaintiff's claims. Furthermore, the coverage percentage of the listed elements in the reasoning is useful for identifying potential gaps in the legal argumentation.

Case	Claim	%C	Arg	%C	Evid	%C	Decision
0253487-55.2020	8	75%	9	78%	5	100%	PG
0006309-46.2010	6	100%	6	84%	5	100%	G
0130337-08.2018	8	75%	9	78%	8	88%	G
0154724-53.2019	5	60%	6	84%	7	72%	D
0243585-78.2020	10	80%	9	78%	8	100%	G

Table 5. Statistics of claims, arguments, and evidence extracted from the legal documents of each judicial case, and the coverage percentage (%C) of each of these elements in the legal reasoning

5 Conclusion

This work provides a comprehensive and technical overview of an advanced pipeline for the automated generation of judicial reasoning, integrating cutting-edge Artificial Intelligence technologies with the specific requirements of Brazilian law. The proposed system not only optimizes judicial processes but also sets a new standard for quality, consistency, and reliability in decisions, representing a significant advancement in the legal field.

The results demonstrate the pipeline's potential to substantially reduce the time required for drafting judicial decisions while improving the accuracy of legal reasoning. This approach enables judges and legal professionals to focus on more complex and strategic analyses, fostering a more agile and effective justice system. For instance, by explicitly linking evidence and claims to legal reasoning, the framework facilitates audits, reviews, and enhances the transparency of decision-making processes. Despite these significant contributions, critical challenges remain for broader implementation of this technology, such as the need for human oversight, ensuring explainability, and adapting the system to different legal frameworks. Addressing these issues is crucial to safeguarding the principles of fairness and justice.

Future developments include integrating the pipeline with Ontologies and Knowledge Graphs, enabling greater traceability and explainability of judicial decisions, as well as the potential application of the framework in international legal contexts. This evolution paves the way for a more robust judicial system that balances technological innovation with the fundamental values of law. Ultimately, this work highlights how Artificial Intelligence can be a transformative ally in legal practice when applied ethically, responsibly, and transparently. It invites reflection: how far can technology advance the judiciary without compromising its essential mission of delivering justice?

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